Tab 1

DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 —NOTES

Submitted By: Gunjan

Domain: Cybersecurity / SOC Analyst

Platform: TryHackMe

Submission Date: 15 October 2026

MITRE ATT&CK

1. What is MITRE ATT&CK?

MITRE ATT&CK is a knowledge base that contains information about how attackers behave during a cyber attack. It is based on real-world observations of different attack methods used by threat actors.

ATT&CK stands for **Adversarial Tactics, Techniques, and Common Knowledge**.

It helps security teams understand what an attacker is trying to do and what techniques they may use during different stages of an attack.

In simple words

MITRE ATT&CK is like a reference guide for attackers' behaviour. It helps a SOC analyst connect an alert or suspicious activity with a known attacker technique.

---

2. Why is MITRE ATT&CK important for SOC analysts?

MITRE ATT&CK is useful for SOC analysts because it gives them a common way to understand and classify suspicious activity.

Some important uses are:

- **Alert investigation:** An analyst can map a security alert to an ATT&CK technique.
- **Threat detection:** It helps security teams understand what attacker behaviour they should look for.
- **Incident investigation:** Analysts can use ATT&CK to understand what an attacker may have done after gaining access.
- **Threat hunting:** Analysts can search logs for behaviours related to specific ATT&CK techniques.
- **Incident response:** It helps in understanding the possible actions taken by an attacker.

- **Security improvement:** Organizations can identify areas where their current detection rules may be weak.

For example, if a SOC analyst sees suspicious PowerShell activity on a Windows system, they can investigate it and map it to the relevant MITRE ATT&CK technique.

---

3. How does MITRE ATT&CK help in threat detection?

MITRE ATT&CK focuses on **attacker behaviour**, rather than only looking at malware names or specific indicators.

For example, an attacker might use:

- PowerShell to execute commands
- Phishing to trick a user
- Credential dumping to obtain passwords
- Network scanning to discover systems
- Scheduled tasks to maintain persistence

These behaviours can be mapped to ATT&CK techniques.

This makes it easier for a SOC team to understand what is happening and decide what logs, alerts, or detection rules should be checked.

---

4. Example of MITRE ATT&CK in a SOC investigation

Suppose a SOC analyst receives an alert showing that:

> `powershell.exe` was executed with a suspicious command.

The analyst would not immediately assume that the system is compromised.

They could investigate:

1. Who started the PowerShell process?

2. What command was executed?
3. Which user account was involved?
4. What process started PowerShell?
5. Was PowerShell downloading or executing anything?
6. Were there other suspicious events around the same time?
7. Does the activity match a known MITRE ATT&CK technique?

If the behaviour matches PowerShell execution, the analyst can map it to the relevant ATT&CK technique under the **Execution** tactic.

This gives the analyst more context about what the attacker may be trying to achieve.

---

5. Main benefits of MITRE ATT&CK

Benefit	*How it helps*
Alert investigation	*Helps classify suspicious activity*
Threat hunting	*Gives analysts behaviours to search for*
Detection engineering	*Helps create detection rules*
Incident response	*Helps understand attacker activity*
Threat intelligence	*Provides information about known attacker behaviours*
Security testing	*Helps identify gaps in security controls*

---

6. MITRE ATT&CK and SOC

For a SOC analyst, MITRE ATT&CK is mainly useful as a framework for connecting **security events with attacker behaviour**.

For example:

Log/Alert → Investigation → Attacker Behaviour → ATT&CK Technique → Tactic → Response

This can help an analyst move from simply seeing an alert to understanding what might actually be happening on the system.

---

7. My Understanding

Before learning about MITRE ATT&CK, I mainly thought of security alerts as individual events that needed to be checked.

After learning about ATT&CK, I understand that an alert can also give information about the **behaviour of an attacker**.

For example, a suspicious PowerShell command is not just a PowerShell event. Depending on the context, it can be part of a larger attack where the attacker is trying to execute commands, download something, or move further into the environment.

So, I think MITRE ATT&CK is useful for a SOC analyst because it provides a structured way to understand and investigate attacker behaviour.